CellForge: Language-Model Agents as Direct Authors of Neural Networks

Authoring, breeding and evolving small gradient-free cell graphs

CellForge project*

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Abstract

We ask whether large language model (LLM) agents can build useful neural computation by *directly writing* weights, biases and wiring, with no gradient-based fitting of the resulting artifact, and whether small authored cells can then be recombined by breeding and evolution. CellForge provides a recursive graph representation (primitive cells composed into nested, shareable bundles with optional conditional dispatch), atomic validated transactions, a typed `import_def` crossover, an append-only hash-chained ledger, and a final evaluation on a separate machine with freshly generated cases. On variable-length digit strings, four independently authored artifacts—including one written with no metric feedback—read in-distribution final cases perfectly (exact match 1.000, $n = 2000$); under distribution shift the best reads 73.8% of strings in an unseen font (72.3% under size shift; the others 38–57%), against 10.0% for a gradient-trained slot-MLP control (83.8% in distribution) and 0% for random direct edits. Authored character language models reach 0.678–0.681 bits per character on a synthetic grammar, but authored and control models score near chance (0.224–0.266) on public multiple-choice probes. On ARC-AGI-2 public-training development tasks, authored graph cells solve 3/200 tasks and 4/200 with one object gene; breeding, mutation and random search mostly tie, and an evolved cellular-automaton control solves 0/40. These are development results on exposed data, from single small runs, with no private-set score. The system is not a language model and nothing here bears on general intelligence.

1 Introduction

Neural networks are almost always *fitted*: an optimizer adjusts parameters to reduce a loss on data. LLMs, however, can read images, reason about mechanisms and write precise numerical specifications. This suggests a different construction process: an LLM *authors* a network the way an engineer writes a program, observes its failures and revises it. CellForge is an experimental system for testing this idea, and for testing whether small authored modules (*cells*), grouped into *bundles* and *circuits*, can be joined by *breeding* (typed crossover of imported definitions) and *evolution* (discrete mutation and selection). Everything is pure in the sense of the project owner's rule: no gradients of any kind, including fitness-weighted gradient estimators, are used in primary arms. The target hardware is low power: two CPU cores and 4 GB of RAM.

Our primary question is whether LLM agents can discover useful neural computations, their recursive organization and better ways of developing them, using evidence-backed memory and

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direct numerical and structural edits, producing an offline artifact that improves on independent evaluation.

How results are validated. Development data, reference anchors and all diagnostics are author-visible and count as development feedback. Final cases are rendered on a separate server by an `evaluator` account with a fresh secret seed per evaluation, and only aggregates leave the machine. Every run records hashes, ledger entries and a receipt file under `runs/`; each claim below names its receipt where one exists. Many later experiments use exposed public development tasks for which no final split exists; those are labelled development results throughout.

Scope and non-goals. The artifacts studied are an offline image-to-text recognizer, small authored byte-level language models, and authored grid-transformation graphs. None has language understanding, dialogue, speech capability or general intelligence. The project owner's long-term aim is more ambitious (a conversational, multimodal system whose computation is constructed rather than fitted); we state it only to explain design choices (modality-agnostic decoder, recursive reusable components) and keep it strictly separate from our claims. Section 6 lists what would have to be demonstrated, in order.

Contributions. (1) A recursive cell/bundle IR with lazy conditional dispatch, content hashing, behaviour-preserving refactors and validity-preserving typed crossover; (2) a transaction and governance design that makes each parameter change traceable to an accepted proposal and separates development feedback from independent evaluation; (3) measurements of LLM-authored artifacts against gradient-trained and random-search controls on digit strings, synthetic and real text, MNIST-1D, public multiple-choice probes, and ARC-AGI-2 development tasks, including several negative results.

2 Related work

Architecture search composes motifs hierarchically [1] and evolves topologies and components [2]; both fit weights by training. Our evidence memory adapts ideas from visual retrieval-augmented generation [3, 4, 5]. Neural cellular automata have been applied to ARC-AGI with a separate gradient-trained model per task [6]; our evolved variant (Section 4.9) replaces gradients by discrete selection and is a different experiment. Tracr compiles hand-written programs into transformer weights without gradient descent [7], the closest prior example of directly constructed weights that we recorded. A recorded report of LLM-seeded genetic search over fixed-width programs on ARC-AGI-2 [8] motivates our validity-preserving genome; we did not independently verify its numbers and do not compare against them. We are not aware of prior work in which LLMs author all weights of a working recognizer without fitting, but a wider prior-art review is required before any novelty claim.

3 System

3.1 Representation

A *cell* computes $h = \sigma(W_1 x + b_1)$, $y = W_2 h + b_2$ over the last axis of its input. A *bundle* is a typed DAG whose nodes are parameter-free operations, parameter tensors, cells, other bundles, or a top-1 `dispatch` among bundles; unselected branches do not execute. The same representation

is used at every depth; definitions may be shared and parameter sharing is explicit. Recursive template references and exponential expansion are rejected at validation. A model has an `encode` bundle (pixels → memory) and a `step` bundle (memory, step index, previous token, state → logits, state), driven by a bounded greedy decoder that stops on the model's own EOS. Structural profiles (ladder, ladder-micro, open) constrain organization and are checked by a validator.

3.2 Construction protocol and maker operator

Makers submit JSON proposals of atomic operations (set parameter by explicit value or by a deterministic expansion of an author-written pattern; put/delete definitions; wrap/unwrap). Proposals are validated against their exact base hash, executed on sample inputs and, for pure refactors, tested for exact behavioural equivalence. Makers may not run code that computes weights from data, may not script search loops, and receive development feedback only through a budgeted `dev-eval` command. A manual-maker operator records development train-pair diagnostics, authored transactions, actor/model provenance, graph hashes and candidate status. Each serializable proposal consumes a budget unit, including invalid, rejected and unchanged edits; transactions are applied to a copy and checked against graph validation, structural profiles and typed runtime fixtures. Three focused tests pass. The operator calls no LLM service and performs no benchmark itself, so a matched non-LLM mutation arm must use the same proposal budget and access. A frozen candidate receives one ledger-accounted final call, consumed even on failure. This is accounting infrastructure, not capability.

3.3 Breeding and evolution

Breeding uses checked `import_def` crossover: a child imports definitions from parents with namespace, shape, ownership and resource validation. The proposed and later implemented genome uses fixed slots with typed ports, bounded parameter shapes and shared parameter identities; a slot may exchange only a compatible definition, and crossover preserves dependency order, declared state and the two-attempt grid interface. Broken wiring should be prevented by construction; static validation remains a final safeguard and does not show task competence. Children commit parent hashes, seeds and donor-per-slot choices, and neutral lineage metadata does not change canonical genotype identity. Primary selection is discrete (μ, λ) population selection with Gaussian coordinate or block mutation; it is not a fitness-weighted gradient estimate. The owner later authorized automated numeric fitness selection and count-fitted seeds (dev-training corpus only), and then prohibited gradients of every kind and excluded the Hedge-style adaptive mixer from all primary arms. Evolved parameters are therefore reported as fitness-selected rather than hand-authored, with donor provenance preserved.

3.4 Governance and evaluation boundary

A hash-chained ledger, graph version store and receipts make each run auditable; reported ledger-entry counts were verified after retrieval. Final cases for the digit task exclude, for strings of length ≥ 4, every string authors ever saw; the unseen-font split uses a font file never present on the authoring machine. Reserved audit data must never be shown to a maker, and schema guards alone cannot prove information isolation.

Artifact	iid	unseen font	size shift	length 9
Authored, iterative A	1.000	**0.738**	**0.723**	1.000
Authored, iterative B	1.000	0.382	0.379	0.000
Authored, one-shot	1.000	0.427	0.453	1.000
Authored, flat	1.000	0.407	0.570	1.000
Gradient slot-MLP, 400	0.057	0.054	0.050	0.000
Gradient slot-MLP, 20k	0.838	0.100	0.419	0.000*
Random edits, 20k evals	0.000	0.000	0.000	0.000

Table 1: Digit strings: exact-match rate on independent final cases (fresh seeds). Wilson 95% intervals are ± 0.03 or narrower at these n; CER in the project log. *The slot-MLP has eight character slots by construction.

4 Experiments

Unless stated otherwise each experiment is a single run per arm.

4.1 Digit strings

Protocol. 32×128 grayscale, 1–8 digits, DejaVu Sans Mono 20 px, horizontal offset 2–14 px, vertical jitter ±2 px, Gaussian noise $\sigma \leq 0.03$. Final splits: iid ($n = 2000$), unseen font Liberation Mono ($n = 1000$), size 18–22 px ($n = 1000$), length 9 ($n = 500$). From an EOS-constant bootstrap: two independent iterative makers with recursive bundles (25 dev-eval budget each), a flat-graph maker, and a one-shot maker (validation only, no metrics). Controls: a gradient-trained slot-MLP (Adam, 400 or 20,000 training images) and (1+1) random direct-edit hill climbing (25 or 20,000 evaluations), all scored through the same reference evaluator.

Results. Table 1. All four authored artifacts are perfect in distribution, using two or fewer development evaluations each, so the in-distribution task is saturated and does not discriminate arms. Every maker converged on the same mechanism family (vertical alignment, left-edge detection, a fixed 12 px character pitch, hand-typed 15×12 glyph templates), yet small decisions produced large robustness differences: artifact A's normalized agreement score and ±3 px search tolerance generalized best; artifact B hard-coded eight character cells and failed every 9-digit string. LLM authors can construct compact recognizers (about two thousand parameters for artifact A) without gradient fitting, and the one-shot author needed no feedback. The results do *not* show a difference between arms (one run per arm), robustness to real-world text, or superiority over gradient training: the control lacks translation invariance and a gradient-trained convolutional network would likely be perfect in distribution. The templates came from *looking at* development images; we regard the author's visual perception as the channel through which learning happens and report it as such.

4.2 Synthetic-grammar hierarchical mixture-of-experts language models (`tinylm/1`)

The owner judged digit recognition too easy a test of the intended cell→cluster→expert design, so authors built character-level language models with routing and recurrent state doing real work.

Task and arms. Text comes from a private English-like grammar (12 nouns, 12 verbs, agreement that must survive relative clauses, adjectives, prepositions, adverbs, ", and" coordination) held only on the evaluation server. Authors see 2,000 sampled documents and 300 development documents,

Model	bpc	top-1	gramm.	gramm.+novel	experts	params
Ladder L1	0.6796	.810	.966	.788	6	278k
Ladder L2	**0.6776**	.809	**.997**	.884	60	1.69M
Ladder L3	0.6790	.809	.940	.804	4	898k
Open O1	0.6797	.809	.955	.839	–	1.31M
Open O2	0.6808	.809	.939	.775	–	1.56M
Gradient MLP (window 8)	0.824	.780	.201	.135	–	74k
Trigram counts	1.383	.617	0	0	–	66k
Uniform	5.000	0	0	0	–	32
Generator entropy: ≤ 0.6734 bpc (summed choice bits, upper bound)						

Table 2: Synthetic-grammar language models: independent final evaluation on 1,000 fresh documents (bootstrap 95% intervals on bpc about ± 0.002). "Gramm." is the fraction of complete generated sentences accepted by the private grammar; "+novel" also requires the sentence to be absent from the authors' training text.

never the grammar. Vocabulary: 32 tokens; the model receives the last eight characters, the position and its own state. Scoring is bits per character (bpc) under teacher forcing on fresh documents, plus grammaticality of 200 free-running samples (temperature 0.8) judged by the private recognizer. Three authors used an enforced ladder profile (cells inside clusters, clusters inside experts, a router driving lazy top-1 dispatch among at least two experts), two an open profile. Controls fitted from data on the same 2,000 documents: uniform, n-gram counts and a gradient-trained MLP over the same eight-character window (Adam, 74k parameters).

Results. Table 2. All five authored models reach 0.678–0.681 bpc, within 0.004 bpc (under 1%) of the generator's entropy, with 94–99.7% grammatical generations, against 0.824 bpc and 20% for the gradient MLP. The authored models keep singular/plural state across nested clauses in a stack held in recurrent state; the fixed-window MLP cannot. The best model (L2) has 60 routed experts, one per grammatical situation, and only 3 of 60 sampled sentences violate the grammar. The task is exhausted: every author read roughly 40 documents, wrote the closed grammar down, hand-tallied rates and compiled it into tables and state machines. This shows the machinery (state, recursion tracking, dispatch among tens of experts) works end to end at 0.3–1.7M authored parameters. It does *not* show that the ladder helps: an open, unrouted model (O1) matched the ladder with three feedback runs against five to eleven, and ladder authors were required to use routing the task does not need. Nor does it show that authoring scales to data whose generator cannot be read off a few dozen samples. Author-reported cost: 176–231k tokens per run.

4.3 Real-text mixture of experts, delegation and failure-informed mutation

The next step moved to real text, where no closed grammar exists. In a delegated real-text model (hash `82ebe5e5...`), three isolated mutations reused settings from earlier failed proposals. The first worsened development bpc from 2.4748 to 2.4826. A trigram-donor change scored 2.4746 bpc, effectively tied with its parent; an after-period-space specialist scored 2.4743 bpc. These comparisons used the 165-document development set. The specialist was accepted on a separate eight-expert branch; its reference-anchor bpc was 2.46152 versus 2.46236 for the seven-expert parent over 155,239 tokens, with identical 0.5220 top-1 accuracy. Both checks are development feedback and establish no final or frontier-benchmark gain; the main branch remains seven-expert.

Artifact	Final accuracy	Wilson 95% interval
Authored A1	0.4655	[0.4437, 0.4874]
Authored A2	0.4360	[0.4144, 0.4578]
Authored A3	0.3410	[0.3205, 0.3621]
Authored two-parent crossover	0.5295	[0.5076, 0.5513]
Authored three-parent crossover	0.5360	[0.5141, 0.5578]
Gradient logistic	0.3355	[0.3151, 0.3565]
Gradient MLP, 100 hidden units	0.5940	[0.5723, 0.6153]
Gradient MLP, 500 hidden units	0.5995	[0.5779, 0.6208]

Table 3: MNIST-1D independent final accuracy; $n = 2000$ per artifact. The MLP controls were not tuned to match published MNIST-1D baselines.

A UTF-8 byte lift of the eight-expert text MoE retains authored weights and 225-dimensional recurrent state in one graph with a shared 259-token head. Its inherited experts still use only eight projected bytes. A 387-token ASCII compatibility probe preserves 208 correct predictions, with loss changing from 2.4798989 bits per character to 2.4803443 bits per byte. This establishes input plumbing, not language competence; three ordinary or Unicode prompts repeatedly emit "too" to the token limit (receipts: `runs/unified-byte`). A directly authored repetition-memory mutation reading a 64-token context has parent loss 2.480344 versus 2.512589 bits per byte on the same 387 bytes; both predict 208 tokens correctly with zero scoring failures, and both answer 0/4 author-visible novel key/value binding prompts exactly. The mutation was rejected for promotion (`runs/parallel-reasoning/DECISION.json`); this small development probe is not a held-out reasoning evaluation.

4.4 MNIST-1D direct authoring and crossover

MNIST-1D provides a procedural ten-class task on 40-sample signals. We used 4,000 development-training signals, a separate 1,000-signal development evaluation set and a 1,000-signal reference anchor; independent final evaluation used 2,000 newly generated signals per artifact with an operator-side secret seed. Makers were told not to inspect the generator source, a restriction that was procedural rather than technically enforced.

Round A and controls. Three independent makers started from the same zero-logit bootstrap under harness version 1, wrote numerical filters and classifier wiring without gradient fitting, and used respectively four, five and four development evaluations. Table 3 shows the strongest authored parent exceeding the logistic control but trailing both MLP controls (trained on the same split). Class-specific strengths differ substantially; three runs are insufficient to estimate a typical outcome.

Crossover. A fourth maker imported the independently authored A1 and A2 filter banks into a child graph and authored a combining rule. After five development evaluations the accepted child scored 0.5295 on fresh final signals, an absolute gain of 0.064 over the better parent; it uses about 29.7 million multiply–accumulates over the final set versus 13.3 million for A1 and 16.4 million for A2. This is one selected offspring, not evidence of a general evolutionary advantage. A three-parent extension scored 0.572 on the development set and 0.557 on the reference anchor ($n = 1000$ each), then 0.5360 on a separate fresh-seed final set ($n = 2000$). Its apparent 0.0065 gain over the two-parent child is not reliable: the final evaluations used different seeds, the intervals overlap

substantially, and the third parent raises final-set computation from 29.7 to 39.3 million multiply–accumulates. A version-2 diagnostic plugin reporting class confusion and wrong-prediction margins was validated and activated, but fresh-maker Round B under that harness was never evaluated, so no harness-effect claim is possible. New MNIST-1D jobs were paused under the later document-benchmark priority.

4.5 Frontier-benchmark probes at chance

We adapted character-level models to multiple-choice scoring (`mc-loglik/1`) by summing next-character negative log likelihood over each answer choice, also reporting length-normalized scoring. The vocabulary lowercases text and omits many capitals, digits and symbols, so coverage is imperfect. MMLU test, ARC-Challenge test and HellaSwag validation were staged on the private evaluation server; GPQA was unavailable because access requires dataset authorization.

The first sweep covers a synthetic-grammar authored model and three real-text controls (unigram, trigram, small gradient MLP), each on a fixed 1,000-item subset per benchmark. Length-normalized accuracies span 0.243–0.266 on MMLU, 0.224–0.248 on ARC-Challenge and 0.230–0.264 on HellaSwag. The delegated real-text MoE (`82ebe5e5...`) on 1,000 items each scores 0.241 MMLU, 0.243 ARC-Challenge and 0.241 HellaSwag (Wilson intervals $[0.2155, 0.2685]$, $[0.2174, 0.2705]$, $[0.2155, 0.2685]$; character coverage 0.9147, 0.9624, 0.9889; receipt `runs/bench_results/server-aggregate-2026`). All values sit around the four-choice chance rate of 0.25 and provide no evidence of frontier benchmark competence; they are negative transfer evidence and a test of the scoring interface, not a comparison to frontier language models.

4.6 Byte-level language graph, component specialists and the document hold

Tool-loop probes. The actual byte-language graph (hash `f0b43dbe...`) was asked to generate tool-loop actions on three authored synthetic prompts (one policy step, 48 generated tokens per action). It produced three invalid JSON actions and zero tool calls or responses (`runs/parallel-tools/server-poli`). This language-policy failure is retained alongside the following component results, each a narrow authored algorithm evaluated by the same research team and not evidence of language understanding:

- Typed finite-state graph (`7f810c68...`), `typed-observation-transfer/1`, 128 fresh numeric payloads of width seven (seed 90218, 640 policy evaluations): transfers 128/128 observations versus 0/128 with observations removed, abstains on 128/128 no-request and 128/128 invalid-observation cases and calls on 128/128 missing-observation cases (`runs/parallel-tools/server-typ`).
- Exact binding graph (`e36653df...`, zero parameters): 128/128 server-generated lexical-variation cases (Unicode, missing keys, prefix distractors, duplicates) match, versus 0/128 for a disabled "unavailable" component (a dependency ablation, not a baseline); receipts in `runs/parallel-reasoning/binding`.
- Carry-addition graph (`43c2a56a...`): 63/63 development additions versus 2/63 for an identity control, all five invalid inputs rejected (`runs/parallel-arithmetic/REPORT.json`); its empty source-hash field limits provenance.
- Frozen digit OCR graph (`54e3934a...`) returns correct text for five authored digit strings; the negative control "hello" yields "50115" and blank input yields empty text (`runs/parallel-vision/server-s`).
- Positive-Horn deduction graph (`3e9019e4...`): 120/120 development cases (71 entailed, 49 not entailed) versus 56/120 with rules removed, 120/120 under rule-order permutation (`runs/parallel-deduction/REPORT.json`).

Composition. The action-grammar graph `8868695d...` answers two fresh synthetic digit images ("28", "39") correctly through the actual specialist harness, each with one OCR call and zero invalid actions; with one observation withheld it returns no answer and repeats the call twice, and with the tool disabled it returns no answer with zero accepted calls and two invalid actions (`runs/action-language/harness-development-result.json`). The composed graph `e04777f4...` preserves inherited MoE dispatch and gives the grammar graph-owned priority; it is bit-identical to its parent on one ordinary-input test, and generates OCR text "587" under preservation probes (four focused tests), and succeeds on two further digit images ("88", "86"). A matched 387-byte probe gives parent and composed graphs identical loss (2.480344 bits per byte) and 208 correct tokens. The source-stability flag is false because a batch method was appended during execution; the coordinator reports identical serial-method ASTs and the executed snapshot is retained in `runs/action-language/composed-runtime-snapshot`. A tiled variant `ab3cee4e...` preserving unused inherited arrays passes the strict micro profile with identical 387-byte scores; it retains 81,432 parameter slots, 3,868 nonzero values, 96 cells, 26 bundles, 613 instances and one dispatch, edges rise from 14,327 to 15,618, and a single local run takes 12.097 s versus 9.746 s untiled (about 24% overhead, uncertain). The controlled-English deduction composition `bef61dc6...` calls component `3e9019e4...` in an immutable five-condition harness (`runs/action-deduction/harness-snapshot-v1/result.j` positive, missing-fact and missing-rule cases answer correctly (3/3), each with one call, zero invalid actions and zero specialist errors; withheld observations give no answer and two repeated calls; disabled tools give no answer, zero accepted calls and two invalid actions. Condition latencies are 163.1, 150.6, 144.1, 140.8 and 107.6 seconds. The grammar is bounded unary controlled English, and repeated unavailable calls and high latency remain failures. The coordinator reports 153 full-suite tests passing in 81 seconds; that is regression evidence, not capability.

Document benchmark hold. The priority benchmark is MMLongBench-Doc (repository); this is a research priority, not a claim that it is the hardest benchmark. All five OCR graph families are digits-only, fixed 32×128 raster interfaces with no question conditioning or layout encoding and cannot yet meet its contract; a digit-only proxy would establish nothing. The pinned source has 135 PDFs and 1,082 annotated questions (the README records 1,091; see Section 7). Private staging completed with zero benchmark evaluations. The first split was rejected after two byte-identical PDF groups and one cross-split conflict (`runs/document-crossover/server-stage-v2.json`); canonical grouping gives 133 unique byte groups, provisionally 107 development and 28 held-out document IDs. The related-edition audit reports 67 candidate pairs, nine crossing the split, and 40/133 groups with inadequate extracted text; a later audit groups 117 heuristic families, which does not verify isolation. The split is not frozen. Alphabet-expanded OCR (`8332b0c6...`) recognizes three of six tiny development strings; its 61 literal-glyph checks test the templates, not accuracy. Three OCR crossover children are frozen without selection. The same-item diagnostic (`document-phase-diagnostic/3`) renders in 1.6596 s, completes all 2,125 OCR crops in 133 batches (1.7213 s), then enters language generation at 2.1396 s with a 783,438-byte context and times out at its 120-second limit (`runs/document-crossover/server-phase-diagnostic-v3.json`; the recorded `language_seconds=0` is an unfinished timer). A compact serialization (v4) cuts the context to 185,055 bytes (a 76.4% reduction) yet still times out after 1.7736 s of OCR (`runs/document-crossover/serv` this negative result selects no model. A compiled-runtime backend initially shows about 1.4× step speed; differential qualification (29 tests, 74-fixture injection suite; first NaN failure retained and repaired) reports amortized speedups of about 21–26× and first-request speedups of about 5–7.5× with matching states, logits, tokens and statistics on tested cases (`runs/compiled-runtime`); it is opt-in and is an implementation measurement, not capability. No document score exists.

A dated comparator plan (`docs/FRONTIER_COMPARATOR_PLAN.md`) names endpoints `gpt-6-astra` and `claude-fable-5-1` (verified from vendor documentation on 29 September 2026); it matches ordered 72-DPI page rasters and questions, no paid requests have been made and endpoint access is unverified. The measured language failures motivated `docs/LANGUAGE_CORE_OPTIONS.md` (a trained shared trunk, the no-gradient primary, or a three-size study); the owner rejected gradient descent and the trained-core option, so the option is historical. Document scoring is held until a predeclared plain-text QA gate; a proposed 16/20 coherent-answer threshold is not frozen and is not a goal-completion criterion. General language capability, not transport, is the blocking question.

4.7 Evolution pilots on the byte MoE and count-fitted seeds

Generation-zero calibration. `runs/evolution-language-gen0-v2.json` records the founder `f0b43dbe...` on 334 content bytes at 2.042730829833363 bits per byte, zero failures, 9.81954 s single-thread nice-15 (about 34 content bytes/s on this workload). This is selection fitness on public development-training material with EOS excluded; V1's requested 150–256-byte stories selected no cases, and V2 used 288–350 after a 288–600-byte inventory, without using model outputs. A local founder curriculum receipt (`runs/language-curriculum/FOUNDER_DECISION.json`) on eight authored cases scores its 190 content bytes at 4.0663511910499714 bits per byte; raw copy, recall and QA each fail all two cases and all six generations repeat "too" to the 24-token cap (9.64137 s). These cases are separate from the 334-byte calibration, so the losses are not comparable.

Mini-pilot (null). `runs/language-evolution-minipilot-v1-launch.json` records a single-thread nice-15 run with a 2-GiB address-space limit and 600-second timeout: three generations, each proposing one router-array and one expert-array candidate, giving six GA evaluations and six founder-random controls on paired seeds; each proposal mutates eight nonzero coordinates with relative Gaussian scale 0.05 and survives only on strict development improvement. It completes in 152.67957 s with peak RSS 68,136 KiB (about 66.54 MiB), twelve candidate evaluations, two unique report graphs and a sixteen-entry ledger (`runs/evolution-language-minipilot-v1/result.json`). Both arms score 2.042730829833363 selection bits/byte at generations zero and one and 2.0422575408863954 at generations two and three, choosing the same winner `e4d1e5db...`; report-only loss is unchanged at 2.606790482283684 with zero failures. There is no evolutionary advantage or report improvement, report data are public development material, and seven focused tests pass according to the coordinator. Twelve candidates over 152.68 s do not establish a pure candidate rate. Context/state mutations, zero-coordinate activation and crossover are proposed v2 work.

Count-fitted references and seeds. Side research (`notes/CLAUDE_RESEARCH.md`, `scripts/claude_count_refe` reports byte-level order6 Witten-Bell loss 1.3816 on 165 development documents (89,213 bytes), while founder loss 2.5113 uses only 24 documents (12,894 bytes); the scopes are unmatched and establish no matched gap (Table 4). A feature script fitting on development-training documents except the final 150 (used to select mixture weights) reports 1.3909 count-only and 1.2650 with word context and document cache; a side adaptive mixer reports 1.2222. The mixer is a Hedge/exponentiated-gradient-style update, now prohibited and excluded from all primary arms. These are exposed development numbers with non-gradient grid selection, not QA or LLM capability. Server parity of the exact order6 Witten-Bell IR export on the first 24 development documents (12,894 bytes) gives IR and reference both 1.3852251087 bits/byte, maximum position difference about 2.27×10^{-13}; the fitted corpus excludes prior report families, so this is not an exact-scope comparison with 1.3816. Guarded cached scoring on 190 development bytes takes 0.473 s for a router mutant including certification/hash/compile and 0.302 s warm, with bit difference about 1.14×10^{-13}.

Scoring scope	bits/byte	Note
Order6 Witten–Bell, 165 dev docs (89,213 bytes)	1.3816	counts; side reference
Count-only, weights from last 150 train docs	1.3909	non-gradient grid
+ word context and document cache	1.2650	exposed dev feedback
Side adaptive mixer	1.2222	excluded from primary arms
Order6 IR vs reference, first 24 docs (12,894 bytes)	1.3852251087	parity check
Founder, 24 docs (12,894 bytes)	2.5113	unmatched scope
Founder, 334 bytes	2.042730829833363	server calibration
Founder, 190 bytes (curriculum)	4.0663511910499714	different cases

Table 4: Count-fitted references and founder losses. Scopes differ, so rows are not a matched comparison; all data are exposed development material.

Static count-seed four-arm smoke. The bounded static WB6 smoke run (two proposals per search arm) has zero acceptances and unchanged winners; fitted-evolution and fitted-random proposals duplicate at this budget. Eight cached/full qualification cases have maximum bit difference zero (39.986 s). The run takes 25.766 s for 24 batch evaluations (0.931 batch/s); six unique graphs including two baselines correspond to 0.233 unique graph/s end to end. Fitted corpus losses on batches A/B are 1.0819295/0.7147698 versus founder 2.271318/1.621583; fitted long-context fitness is worse than founder, and a 0.05 residual improves corpus loss slightly but worsens context and is rejected. Twelve ledger entries verify; CPU seconds and peak RSS were not captured (`runs/wb6-four-arm-static-v1/`). This two-proposal null result is not scaling evidence.

4.8 ARC-AGI-2 development: authored cell library, gene coverage and breeding ties

Protocol and exposure. Grid receipts use `arc-grid/1`: SHA256 task-ID hash order, with residue zero modulo five reserved for audit and nonzero for development. The scorer validates grids of 1–30 cells per dimension with colors 0–9, accepts at most two candidates per test input and computes exact task success (every test input correct within two attempts) with a Wilson interval. The exposed development manifest (DEV200) contains 200 public-training tasks with 222 test inputs; it is already development exposure, and relabeling a subset as audit does not make it untouched. Historical receipts state that the residue-zero audit was reserved, but later server access cannot be certified from local metadata. Two attempts must be produced by graph-owned selection, not host-side expert ranking. No official ARC-AGI-2 or tool-use score is reported, and the BFCL V4 agentic category needs a search backend not configured here.

First specialists. The `grid20/1` graph routes among 20 identity/transpose and cyclic-recoloring experts using demonstration reconstruction error; it and its fixed-identity control solve 0/200 (no invalid outputs, Wilson $[0, 0.0188]$). The `grid-associative/1` mutation combines eight dihedral experts with demonstration-inferred color associations in graph state and solves 3/200 (1.5%, Wilson $[0.0051, 0.0432]$) versus 0/200 with inferred recoloring disabled (receipts `runs/grid-associative/DECISION.json`, `runs/grid20/DECISION.json`). Fixed two-gene concatenation measures pair coverage but is not a top-two router.

Coverage and curriculum. The graph-gene operator measures 28 individual genes and three materialized pairs on DEV200; every pair solves 3/200, with no breeding gain. Pair throughput is about 51 tasks/s and 0.0195 CPU s/task, with whole-process high-water RSS 87,384,064 bytes. The

dense curriculum has eight families with 200 selection and 100 validation tasks each (2400 distinct original fingerprints): authored selectors solve the five geometry/color families, adding the bbox gene solves crop, and both portfolios fail tiling and local fill. All four graphs solve the same 3/200 native tasks; only two native tasks receive a train-only color-family annotation (both solved), 198 are unclassified (one solved), so family transfer rates are unavailable. The selector is authored, not learned. Whole-process RSS is 168.3 MiB with 139.33 s measured scoring; the synthetic generators are CC0-1.0/MIT-0 and competition eligibility is unverified. The `grid-gene-evidence-v1` receipt records 300/300 synthetic successes each for authored fill and count genes but 0/200 native tasks each; the count gene has one demonstration-consistent native task whose query fails; portfolios stay 3/200. The largest-four component gene passes 100/100 synthetic but solves 1/200 native with zero fully consistent tasks (about 14 tasks/s, 140 MiB; `runs/largest4-evidence-v1/REPORT.md`). Dense synthetic success has not supplied native transfer.

Separate co-researcher library. A co-researcher's 57-cell host-enumerated depth-two library reports 53/1000 public training solves, 65 demo-consistent tasks and 12 false positives (18.4615% conditional on consistency, 1.2% of all training tasks), and a once-scored public evaluation result of 0/120 with no consistent chains. A later version progresses from 80/1000 training solves (v2) to 95/1000 (v3) with v3 evaluation still 0/120 and 12 false positives reported. These are read-only external receipts on exposed evaluation tasks, not a fresh final evaluation or CellForge graph result. Full training access also potentially exposes the modulo-five audit for a combined library despite the operator's zero audit calls. Source imports and v2 filenames do not pin the receipt-generating run; worker-row means are 1.52686 s training and 1.805416 s evaluation, not a verified 0.5 s CPU budget claim, and RSS is unavailable. Its across-query attempt criterion differs from per-input acceptance, so counts are not a matched comparison. Prize-winning odds are assessed qualitatively as very low; realistic deliverables are a valid open entry and an honest paper.

Single-gene and composition measurements. Table 5 lists authored genes (synthetic denominators are original generated tasks; native means DEV200 unless noted). Each row is one frozen graph with verified ledgers and no gradients, public evaluation, audit or final access. The object-association gain was identified by a post-hoc frozen-model observer, complete paired regression attribution is absent, and the child adds operations without matched compute; the 16-step variant's paired check against the unchanged portfolio verifies one gained task and no losses across 200 tasks, and full pairing against 64 steps is unavailable (aggregate receipt). The two scaling genes' three disjoint successes were already identified from development demonstrations outside DEV200, so this is not new independent transfer. The direction router is an explicit maker-authored correction: a scalar lookup had matched the context graph on all 24 selection tasks with all 97 wrong pixels at unknown keys (94 border pixels, none in conflict with learned keys), and the authored router compares five hypotheses (identity and four one-pixel neighbor copies); authoring cost was not measured. A conditional-state v1 prototype without cells and the zero-cell generic version remain historical.

Training-only diagnostics and coverage census. On DEV200, demonstration-only diagnostics show associative-nine with 2/200 consistent tasks, 57 observed shape mismatches and 141 pixel-rule/routing failures; conditional state has zero consistent, 57 shape and 143 failures (17.014 s, 60.28 MiB, four ledger entries). These categories do not separate missing genes from inadequate routing. A structural census found 57 tasks with shape-changing demonstrations; a bank of identity, square tiling, pixel repetition at factors 2 and 3, and bounding crops explained no task exactly,

Graph / gene	Synthetic	Native	Cost and note
Conditional state v2 (2 rank-cell params, 1002 state dims, 1000 adjacency rules)	300/300 vs 0/300 (updates off)	1/200 vs 0/200; none fully consistent	117 vs 115 tasks/s; 49.3 MiB; 12.03 s per run
Adjacency-then-fill serial child (195 slots, 1010 state)	300/300 vs 0/300 for either part	3/200 for all; 2 consistent, 0 false positives	46.88 vs 52.04 and 50.40 tasks/s; 30.97/27.26/28.17 s; 55.4 MiB; 57 slot nodes, 1650 expanded
Direction router (2 ranking params, 6 state entries, 207 edges)	200/200 selection, 100/100 validation vs 0	0/200; none consistent	142.17 vs 147.06 tasks/s; 55.48 vs 55.64 MiB
Object association child (+11 params)	300/300 vs 0/300	4/200 vs 3/200	58.37 to 9.82 tasks/s; 61.27 to 150.95 MiB
Same, 16-step variant (174 params)	300/300	4/200; 3 consistent, 0 consistent wrong	9.82 to 12.49 tasks/s; 0.10180 to 0.08006 CPU s/task; 150.95 to 109.32 MiB; 1564 vs 3004 edges, 205 vs 589 constants
Pixel scaling factor 2 / 3 (31 params each)	300/300 each	2/789 and 1/789 (expanded training)	403.23 / 393.26 tasks/s; 0.002481 / 0.002544 CPU s/task; 21.76 s; 76040 KiB
Graph-owned scaling selection (64 params)	300/300 vs 100/300 (updates off)	3/789 vs 0/789	182.42 tasks/s; 0.005483 CPU s/task; 47.72 s; 70896 KiB
Expanded active-slot graph (7 slots, 273 params, state 1515)	110/120 vs 70/120 (2 specialists active)	not scored	14.44 vs 44.31 tasks/s; 44.31 s wall; 103064 KiB; 40 paired gains, 0 losses
Homologous five-gene bank recombination (25 combos, 206 params)	child 10/10 vs 5/10 per complementary parent	not scored	14.12 tasks/s; 7.20 s; 113188 KiB

Table 5: Authored ARC genes and compositions on exposed development data. Native figures never exceed the 3/200–4/200 range on DEV200; the object gene's extra native task is exposed development only.

and one color association added only two tasks the associative portfolio already reconstructed (0.35 s, 38.11 MiB). Extending the 17-rule census to 789 public-training tasks (nonzero SHA256 modulo 5; 211 residue-zero tasks excluded) gave seven exact matches and 13 with color association, two pixel-repetition tasks at factor 2 and one at factor 3 outside DEV200, no object-removal match, and no unique hole-filling match (5.68 s, 53596 KiB). These motivated the scaling genes; they are demonstration matches, not query solves. Derived cells must retain expanded-exposure provenance.

Matched population pilots. Table 6 collects every crossover, mutation and random-search comparison. Each is a single seed or a few seeds with four to sixteen proposals on exposed synthetic families, with winners frozen before reporting. Physical evaluation counts match across arms but CPU cost does not. No pilot shows a crossover-specific advantage. Details that change interpretation. In the object-bank pilot paired native receipts confirm one gained task and no losses versus either control, but crossover kept object and adjacency genes while random kept crop and adjacency, so equal totals conceal different niches. In the seed replication all 12 crossover proposals failed to combine object and adjacency, both founders died at generation 3 and two populations were extinct after generation 4; early founder loss is a hypothesis, not an explanation, and pure crossover can only reach alleles in living parents. The age-8 run exceeds the pilot horizon and does not support early founder death as the repair. In the seven-gene pilot all four crossover donor draws were $[0, 0]$ (copying one parent), so the complementary child was never evaluated; crossover and mutation share one winner and one report, so these are not independent. In the separate-streams

run distinct winners retained repeat2/repeat3, crop/repeat2 and repeat2/fill, but stream, budget and seed changed together. In the seven-slot run founders scored 14 and 18 on selection, crossover first reached 22 at proposal 10 and mutation at 9, tiling failed in every arm and random also missed object tasks. In the context-projection pilot nine binary key-projection slots were varied with the center retained and all arms kept the orthogonal-cross founder; the graph stores 23 parameter slots. Population mortality and sharing were not separately ablated (shared graphs store 195 slots and 1010 state dimensions), the matched-v1 age/tie rules also changed, and the population-v2 policy lacks a native file-hash manifest, so external byte identity is not asserted.

Seven-slot capacity and breeding interfaces. A finite two-slot diagnosis evaluated all 49 layouts (24 selection tasks, 70 demonstrations, 3430 predictor calls; 285.20 s, 190632 KiB, 101 ledger entries): the demonstration-selected winner scored 14/24, post-hoc best single layout 14/24 and the query-oracle union 22/24 (failing both tiling examples); the union is not a deployable score, and one of two scaling examples per factor tests oversized-query identity fallback. These bounds motivated slot/wiring capacity rather than more search in a fixed space. The shared bank then grew to seven authored genes (49 two-slot combinations valid at 268 stored parameters and state 1010; homologous crossover preserves factor 2 and 3 from complementary parents; checks on 12 crossover and 12 mutation children; five tests), and crossover and one-slot uniform mutation operate on the expanded genome (16 reproducible crossover children and 16 mutations, tamper rejection, four children and eight ledger entries); no offspring fitness was measured for these interfaces. A bounded exact 3-by-3 context memory (23 parameters, 3602 state dimensions, four tests) is frozen but unbenchmarked.

4.9 Evolved cellular-automaton negative control

A co-researcher's scratch numpy experiments (not reproduced here; scripts are outside the repository) evolved tiny neural cellular automata with Gaussian mutation and discrete selection on ARC-AGI-2 public-training same-shape tasks, matched by random and identity controls. A 1488-parameter/T10 variant solved 0/40 tasks after 3000 evaluations per task; synthetic swap loss improved without a solve at 6000 evaluations. A 414-parameter/T3 variant fit synthetic training after 8000–14000 evaluations (1200 evaluations/s on a different six-by-six workload, not attributable to CellForge), with held-out success in one of three configurations, then scored 0/40 training and 0/40 test at 8000 evaluations per task with random and identity both zero; lower pixel loss is not task success. A 313-task ten-core run outside the two-core budget was initially reported pending, then corrected: it stopped at 30/313 with zero solves and is not a benchmark. Raw-weight evolution therefore solves no real task at these budgets, consistent with the need for good seeds noted in related work. The published gradient-trained NCA solves 23 of 172 feasible original-ARC tasks [6]; its GPU timing does not establish CPU performance or no-gradient feasibility. On CellForge hardware, a 1488-parameter/T2 graph passes 5/5 synthetic 30-by-30 identity probes at 93.23 adapter calls/s with 38,268,928-byte peak RSS, and a shallow residual graph (`c985a1ed...`, 414 parameters, 22 microcells, three iterations) satisfies the ladder-micro profile (eight combined and three focused tests) and returns 5/5 at 69.1217 two-phase calls/s with peak RSS 42,491,904 bytes (about 40.5 MiB), single thread and nice 15 (`runs/nca-shallow-v1/server-identity.json`). These are demo/query adapter calls, not ARC fitness; architecture, iteration count and dispatch differ, so parameter count does not explain the speed gap, and no ARC score, evolution or generalization is established.

4.10 Offline submission interface

An offline CPU submission adapter and source bundler have eight and three focused tests; bundling includes transitive dependencies and hash-manifest preflight. A server smoke test on two tasks and three queries takes 0.14686 s at 48,603,136-byte RSS and matches local JSON exactly (`runs/arc-cpu-package/REA`
This establishes interface execution, not Kaggle eligibility or task accuracy.

5 Discussion

What worked. Direct authoring produced working recognizers and compact language machinery without gradients: digit strings read perfectly in distribution, with one artifact retaining 73.8% under an unseen font, and authored synthetic-grammar models reached within 0.004 bpc of the generator entropy. Typed crossover retained parental building blocks on tiny probes and lifted MNIST-1D accuracy by 0.064 in one selected child. Narrow authored components (typed transfer, binding, deduction, addition, action grammar) execute exactly within their grammars, and a compiled backend gives large verified speedups.

What did not. Authored and control character models sit at chance on public multiple-choice probes; the byte language graph emits invalid actions and repeats tokens; document evaluation is blocked on language capability and on an unresolved related-edition overlap; evolution from authored founders tied random search in the language mini-pilot and in most ARC pilots; dense synthetic success (300/300) did not transfer to DEV200, where every portfolio remains at 3/200–4/200; and evolved NCAs solve 0/40. The gradient MLP beat the best MNIST-1D authored model.

Honest limits. Final evaluations exist only for the digit, synthetic-grammar and MNIST-1D experiments; all ARC, byte-language and document results are development results on exposed public data. Runs are single small runs (one per arm in most cases), confidence intervals are wide, and the two-task gaps in Table 6 are not evidence. Authors and orchestrator share an operating-system account locally, and the orchestrator has administrative access to the evaluation server, so the final-evaluation boundary is physical rather than adversarially audited. Token usage is self-reported; the digit task used a single font family in development; the visual-retrieval memory arm and the MNIST-1D harness-version comparison were never run; the MNIST-1D crossover has one independently tested child and no random-crossover control, so its advantage may reflect selection and extra computation; a co-researcher's library results and scratch NCA numbers are not reproduced by us. The authored models transfer competence from the author LLM's pretraining and must be reported as such. There is no private-set score, no official ARC-AGI-2 or BFCL score, the system is not an LLM, and no claim of general intelligence or frontier superiority is made.

Design-only pretrained composition. A System 1/System 2 proposal (`docs/SYSTEM12_DESIGN.md`) would compose pretrained models (a gradient-trained 0.6B-parameter decision-head system and a pretrained generative candidate) with evolved routing. It would import pretrained capability, is separate from the no-gradient line, and was not executed: no model was downloaded, and the approximately 4.34GB required files exceed the reported 3.5GB free disk.

6 Roadmap beyond the current results

The path toward the long-term aim requires, in order: robust alphanumeric recognition across unseen font families; real-world text lines competitive with a parameter-matched gradient-trained CNN (the critical go/no-go test, since authoring effort scales with case diversity rather than data); constructed sequence priors; measured component reuse across visual tasks; constructed algorithmic reasoning with length generalization; and only then language. The largest risks are scale (capable language models have 10^8–10^{12} parameters, far beyond hand authoring, pushing toward weight *generators*) and provenance of linguistic competence. For the ARC line the owner's priorities are a valid open entry, stronger authored object, recoloring, counting and path cells tested on dense generators, and matched ablations of breeding mechanisms.

7 Reproducibility

Each result names its receipt under `runs/`, with graph and source hashes recorded in the hash-chained ledger; ledger counts were verified after retrieval. Fonts are not shipped (`CELLFORGE_FONT_DIR`), the final-evaluation server holds the private grammar and secret seeds, and the document PDFs are pinned but not redistributed. The pinned document source has 1,082 annotated questions while the README states 1,091; we report the pinned count and have not resolved the difference. This manuscript is built with `latexmk` on the evaluation server (TeX Live); the `paper/check_numbers.py` script lists numeric tokens of this file that are absent from the original draft and the research notes.

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Pilot	Synthetic report (crossover / mutation / random)	Native DEV200 and ledger	Cost
One-slot maker (`grid-maker-slot-v2`, seed 662021)	maker and control pick same fill gene and child; 600/800 to 700/800 (reconstruction 0.75 to 0.875)	3/200; 2400 demo-pair scorings per arm	172.02 s; 75,032 KiB
`grid-population-pilot-v2` (4 logical attempts)	120/180 each	3/200 each; 29 ledger entries; 9 unique genomes; cache hits 3/0/2	150.57 s; 73,804 KiB
`grid-population-matched-v1` (6 physical evals, 3120 calls, 18 evals total)	120/180, 140/180, 140/180	3/200 each; 36 ledger entries	228.63 s; 70,388 KiB
Context-projection (4 founders, 8 proposals, 24 tasks)	1/24 development, 0/24 report for all arms, 95.83% pixel accuracy	0 accepted proposals; 41 ledger entries	0.654–0.684 evals/s; 59.39 s; 123.54 MiB
Object bank (2 founders + 4 proposals, 348 calls per arm, 206 params, 1010 state)	14/20, 12/20, 14/20	4/200, 3/200, 3/200; 36 ledger entries	CPU 18.785, 9.704, 13.099 s; 11.59, 46.94, 47.15 tasks/s; 155.77 s; 140.66 MiB
Seed replication (3 seeds, 20 tasks, 18 evals, 1044 calls)	12/20 every run; mutation 14/20 throughout; random 14, 12, 14	not scored; 36-entry ledgers	CPU 38.675, 54.880, 45.086 s
Finite lifetime (max age 3 to 8, same seeds)	crossover 12/20; mutation 14,14,14 to 14,12,14; random 14,12,14	not scored	18 evals, 1044 calls
Seven-gene (seed 772081; 6 graphs, 420 calls)	12/24, 12/24, 14/24	not scored; 36 ledger entries	CPU 8.171, 8.262, 8.824 s; 0.734, 0.726, 0.680 evals/s; 30.34 s; 114324 KiB
Separate streams, 16 proposals (seed 772101; 18 evals, 1260 calls)	14/24 each (tie)	not scored; 108 ledger entries	CPU 26.575, 39.178, 30.947 s; 105.76 s; 152816 KiB
Seven-slot (seed 772201; 18 graphs, 1260 calls)	22/24, 22/24, 20/24	not scored; 108 ledger entries	CPU 89.416, 139.494, 118.018 s; 11.77, 11.70, 36.92 tasks/s; 372.55 s; 167512 KiB

Table 6: Breeding, mutation and random-search pilots on the ARC gene banks. Mutation and random figures follow the crossover column order. Ties dominate; the largest gap (22/24 vs 20/24) is a single-seed two-task difference.